# INTERNATIONAL UNIVERSITY – VNU-HCMC · SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

# IT171IU · Statistical Learning

## Part 1: Checkpoints

### _Checkpoint 1:

**Checkpoint — Simple linear regression**

1. Write the fitted equation for `medv` on `lstat` and interpret the slope in words.
2. What is the difference between the two intervals returned by `conf_int(alpha=0.05)` and `conf_int(obs=True, alpha=0.05)`?
3. Which feature of the residual plot suggests that the relationship is not linear?

1. Fitted equation and slope interpretation:
Equation: medv = 34.55 - 0.95 x lstat
Interpretation: For every 1% increase in the percentage of lower status households (lstat), the average median house value (medv) is expected to decrease by approximately 0.95 units (or $950).
2. Difference between the two intervals:
conf_int(alpha=0.05) produces a confidence interval, which quantifies the uncertainty about the average response (the mean house value) at a specific lstat level.
conf_int(obs=True, alpha=0.05) produces a prediction interval for a single new observation. This interval is always wider because it must account for both the uncertainty of the average and the irreducible error of an individual neighborhood.
3. Feature indicating non-linearity:
The residual plot (fitted values vs. residuals) displays a distinct U-shaped curve (a discernible pattern) rather than points randomly scattered around the horizontal zero line. This clear pattern in the residuals suggests that the true relationship between the predictor and response is not strictly linear.

markdown

### _Checkpoint 2:

**Checkpoint — Multiple regression**

1. Why is `age` significant in the simple model `medv ~ lstat + age` but not in the model with all predictors?
2. Which predictors have a VIF above 5? What does this mean?
3. What does the `anova_lm()` table tell you about the quadratic term in `lstat`?

1. Why age loses significance in the full model:
In the simple model medv ~ lstat + age, the age variable absorbs the effects of other missing variables it is correlated with (omitted variable bias). Once all predictors are included, variables like distance to employment centers (dis) or crime rate (crim) explain that variance more accurately, revealing that age itself does not have a significant independent effect on house values when other factors are held constant.
2. Predictors with VIF > 5 and its meaning:
If you run the list comprehension for VIF on the full Boston dataset, tax (property tax rate) and rad (accessibility to radial highways) have VIF values well above 5. A VIF greater than 5 indicates collinearity—meaning these variables are highly correlated with other predictors in the model. This collinearity artificially inflates the standard errors of their coefficient estimates, making it harder to accurately assess their individual effects.
3. What the anova_lm() table tells us:
The anova_lm() function compares the simpler linear model to the larger model containing the quadratic term poly('lstat', degree=2). The resulting large F-statistic (177.28) and a near-zero p-value strongly reject the null hypothesis. This provides clear statistical evidence that adding the quadratic term for lstat significantly improves the model's fit compared to using only the linear term.

markdown

# Part 2: Exercises:

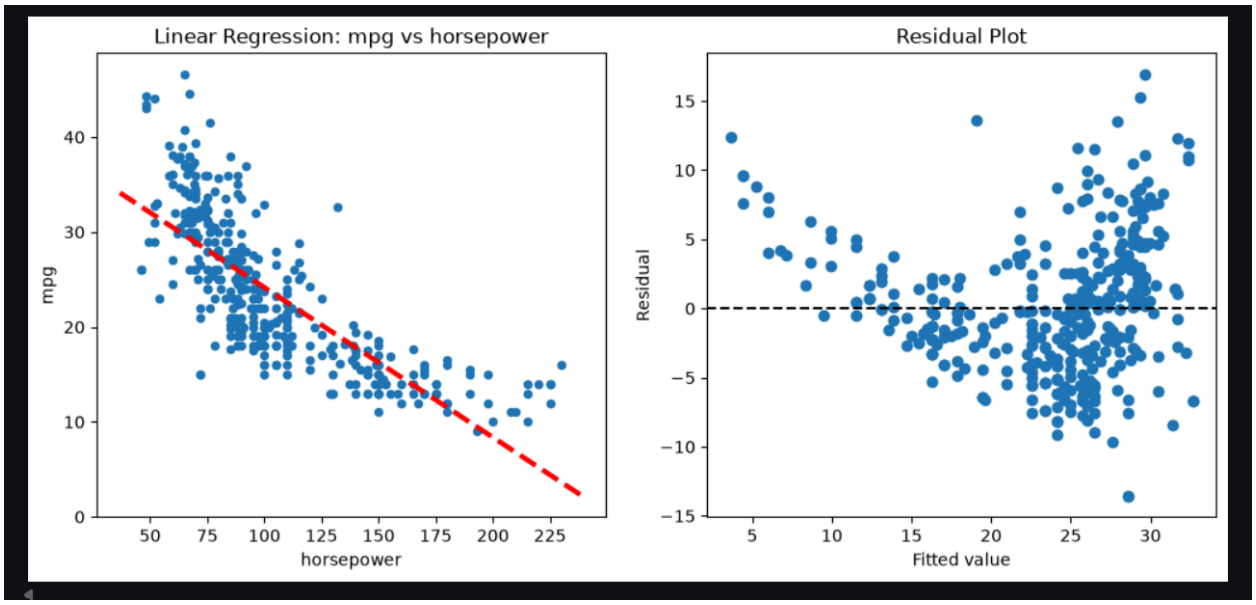
## Exercise 1 – Simple regression on Auto

### Key Numbers:

- **Model Fit:** $R^2 \approx 0.606$**,** p-value for h**orsepower** $\approx 0$**,** coefficient $\approx -0.158$**.**
- **Predictions (hp = 98):**
  - Predicted mpg: 24.47
  - 95% Confidence Interval: $[23.97, 24.96]$
  - 95% Prediction Interval: $[14.81, 34.12]$

### Interpretation:

- **(b) Relationship:** There is a strong, negative, and statistically significant relationship between horsepower and mpg. Higher horsepower yields lower fuel efficiency.
- **(d) Diagnostics:** The residual plot shows a clear U-shape pattern, confirming the relationship is non-linear and the simple linear model is inadequate.



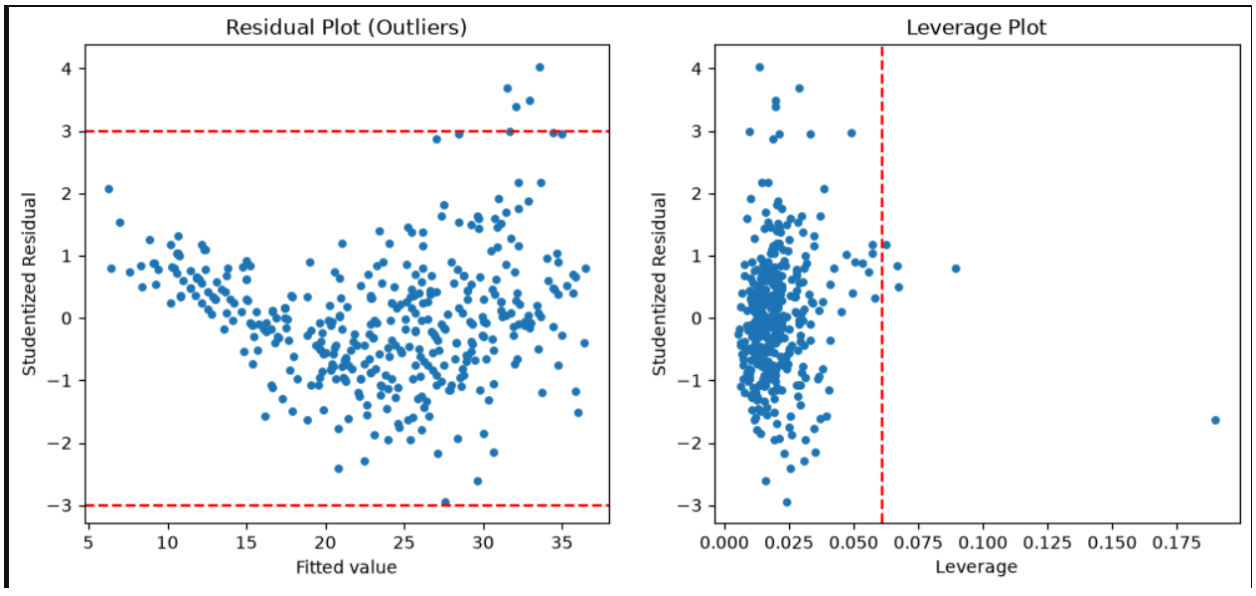
# Exercise 2 – Multiple regression on Auto

## Key Numbers:

- **Correlation:** Strong negative correlations ($-0.75$ to $-0.83$) between mpg and cylinders, displacement, horsepower, weight.
- **P-values:** weight, year, origin are significant ($p \approx 0$).
- **year coefficient:** $\approx 0.75$
- **VIF:** displacement, cylinders, and weight have VIF $> 10$.

## Interpretation:

- **(b) Predictors & 'year':** The positive coefficient for year means that for every one year newer a car is, its fuel efficiency increases by ~0.75 mpg, holding other features constant.
- **(c) Outliers/Leverage:**Outliers exist with studentized residuals $> 3$. Several high-leverage points indicate unusual predictor combinations.
- **(d) Collinearity:** The very high VIF values show severe multicollinearity among the engine and weight-related features.



# Exercise 3: Interactions and transformations

**Key Numbers:**

- **1. Interaction (hp × weight):** p-value $\approx 0$.
- **2. Polynomial (poly(hp, 2)):** p-value for degree 2 $\approx 0$.
- **3. Log (log(hp)):** p-value $\approx 0$, higher $R^2$ than base model.

**Interpretation:**

- All three transformations significantly improve the base model (confirmed by very low p-values and ANOVA).
- Choice**:** The Interaction Model is the best choice because it statistically outperforms the base model and makes strong physical sense (the effect of horsepower on mpg depends heavily on the car's weight).

```
--- Base Model R^2: 0.8083 ---

--- 1. Interaction Model R^2: 0.8574 ---
P-value of interaction term ('horsepower:weight'): 0.000000
ANOVA (Base vs Interaction):
   df_resid          ssr  df_diff      ss_diff           F        Pr(>F)
0     388.0  4565.650523      0.0          NaN         NaN           NaN
1     387.0  3397.738458      1.0  1167.912065  133.024356  1.180727e-26

--- 2. Polynomial Model R^2: 0.8431 ---
P-values of Polynomial model:
intercept                       5.809570e-07
weight                          9.289726e-32
year                            1.767700e-44
poly(horsepower, degree=2)[0]   8.237658e-04
poly(horsepower, degree=2)[1]   1.413402e-18
dtype: float64
ANOVA (Base vs Polynomial):
   df_resid          ssr  df_diff     ss_diff          F        Pr(>F)
0     388.0  4565.650523      0.0         NaN        NaN           NaN
1     387.0  3737.322742      1.0  828.327781  85.773393  1.413402e-18

--- 3. Log Model R^2: 0.8168 ---
P-value of log_hp: 0.000024
```

# Exercise 4 – Qualitative predictors with Carseats

## Key Numbers:

- **Full Model Coefficients:** Price = $-\ 0.054$, US[Yes] = $+\ 1.200$. Urban[Yes] is NOT significant.
- **Small Model 95% CI:**
  - **Price:** $[-\ 0.065, -\ 0.044]$
  - **US[Yes]:** $[0.691, 1.708]$

## Interpretation:

- **(b) Coefficients:** A $1 increase in price decreases local sales by ~54 units. A US store sells ~1200 more car seats than a non-US store on average. Location in an urban area has no significant effect.
- **(c) Equation:** $Sales = 13.04 - 0.054 \times Price - 0.022 \times Urban[Yes] + 1.200 \times US[Yes]$
- **(d) Smaller Model:** Dropping the non-significant Urban predictor yields a simpler model without losing predictive power (ANOVA p-value is high, $R^2$ remains unchanged).

```
--- Full Model Summary ---
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
intercept     13.0435      0.651     20.036      0.000      11.764      14.323
Price         -0.0545      0.005    -10.389      0.000      -0.065      -0.044
Urban[Yes]    -0.0219      0.272     -0.081      0.936      -0.556       0.512
US[Yes]        1.2006      0.259      4.635      0.000       0.691       1.710
==============================================================================

--- Model Comparison ---
Full Model  -> R^2: 0.2393, RSE: 2.4725
Small Model -> R^2: 0.2393, RSE: 2.4694

--- ANOVA (Small vs Full) ---
   df_resid          ssr  df_diff  ss_diff         F    Pr(>F)
0     397.0  2420.874462      0.0      NaN       NaN       NaN
1     396.0  2420.834671      1.0  0.03979  0.006509  0.935739

--- 95% Confidence Intervals (Smaller Model) ---
                  0         1
intercept  11.79032  14.271265
Price      -0.06476  -0.044195
US[Yes]     0.69152   1.707766
```

**Exercise 5 – Simulation**

**Key Numbers:**

- **True Coefficients:** $\beta_0 = -1, \beta_1 = 0.5$.
- **P-value for $x^2$:** $> 0.05$ (Not significant).

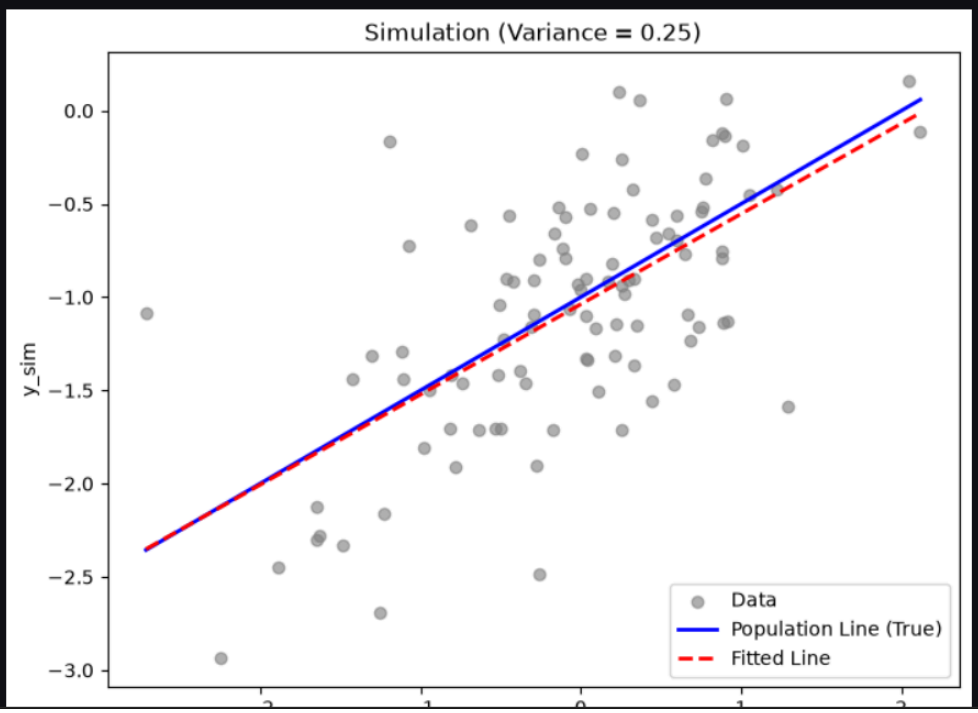
**Interpretation:**

- **(c) Fit:** The least-squares estimates perfectly track the true population line.
- **(d) Quadratic Term:** There is no evidence it improves the fit, which is logically expected since the simulated data is strictly linear.
- **(e) Variance Impact:** With less noise (Var = 0.05), the confidence interval for $\beta_1$ becomes narrower (more precise). With more noise (Var = 1.0), it widens significantly due to higher uncertainty.

```
--- (c) ---
intercept   -1.038013
x            0.484291
dtype: float64

--- (d) P-values when adding x^2 ---
intercept    1.380909e-31
x            1.494913e-11
x2           9.587575e-01
dtype: float64

--- (e) Confidence Intervals for the coefficient of x (beta_1) ---
Var=0.05 (Less noise) : [0.44908133 0.54520308]
Var=0.25 (Original)   : [0.36756536 0.60101667]
Var=1.00 (More noise) : [0.44394622 0.85004259]
```



**Exercise 6 – Collinearity**
**Key Numbers:**

- **Correlation ($x_1, x_2$):** $\approx 0.772$.
- **Combined Model P-values:** $x_1 \approx 0.0028$ (Reject $H_0$), $x_2 \approx 0.259$ (Fail to reject $H_0$).
- **VIF:** $\approx 2.48$.
- **Single Models:** Both $x_1$ and $x_2$ have p-values $\approx 0$ when fitted alone.

**Interpretation:**

- **Contradiction explained:** The results appear contradictory but perfectly illustrate collinearity. Because $x_1$ and $x_2$ are highly correlated, they share variance. The model struggles to isolate $x_2$'s small true effect ($\beta_2 = 0.3$) when $x_1$ is present, incorrectly making $x_2$ look insignificant.

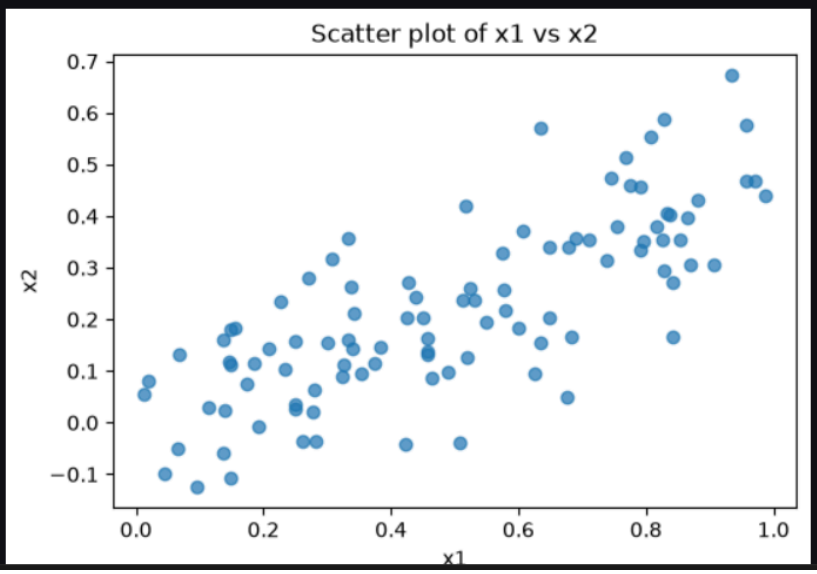
```
--- (b) Correlation between x1 and x2: 0.7723 ---

--- (c) Model with BOTH x1 and x2 ---
intercept    2.747502e-17
x1           2.815785e-03
x2           2.594186e-01
dtype: float64

VIFs:
         vif
x1  2.478223
x2  2.478223

--- (d) Model with x1 ONLY ---
intercept    3.605074e-17
x1           1.367829e-08
dtype: float64

--- Model with x2 ONLY ---
intercept    2.262620e-27
x2           7.429581e-07
dtype: float64
```



# Exercise 7 – Additional data set: diabetes progression

**Key Numbers:**

- $R^2$ **/ RSE:**
  - M1 (bmi): 0.344 / 62.52
  - M2 (bmi, bp, s5): 0.480 / 55.78
  - M3 (All 10): 0.518 / 54.15
- **ANOVA (M2 vs M3):** F-statistic $\approx 4.81$, p-value $\approx 0.000031$.
- **VIF $> 5$:** s1 (59.2), s2 (39.2), s3 (15.4), s5 (10.1).

**Interpretation:**

- **(b) ANOVA:** The near-zero p-value indicates the full 10-predictor model is significantly better than the 3-predictor model.
- **(c) Collinearity:** There is extreme multicollinearity among several blood serum measurements (s1, s2, s3, s5).
- **(d) Most Important Predictors: bmi**, **bp**, and **s5**. These three variables have the lowest p-values ($\approx 0$) in the full model, proving they are the most robust and statistically significant drivers of diabetes progression.

```
(442, 11)
    age  sex   bmi     bp     s1     s2    s3   s4      s5    s6  progression
0  59.0  2.0  32.1  101.0  157.0   93.2  38.0  4.0  4.8598  87.0        151.0
1  48.0  1.0  21.6   87.0  183.0  103.2  70.0  3.0  3.8918  69.0         75.0
2  72.0  2.0  30.5   93.0  156.0   93.6  41.0  4.0  4.6728  85.0        141.0
3  24.0  1.0  25.3   84.0  198.0  131.4  40.0  5.0  4.8903  89.0        206.0
4  50.0  1.0  23.0  101.0  192.0  125.4  52.0  4.0  4.2905  80.0        135.0
M1 (bmi)        -> R^2: 0.3439, RSE: 62.5151
M2 (bmi, bp, s5)-> R^2: 0.4801, RSE: 55.7782
M3 (all 10)     -> R^2: 0.5177, RSE: 54.1542

--- ANOVA (M2 vs M3) ---
   df_resid           ssr  df_diff       ss_diff         F    Pr(>F)
0     438.0  1.362709e+06      0.0           NaN       NaN       NaN
1     431.0  1.263986e+06      7.0  98722.908072  4.809002  0.000031

--- VIF for full model ---
       vif
age   1.22
sex   1.28
bmi   1.51
bp    1.46
s1   59.20
s2   39.19
s3   15.40
...
s6          0.305990
s3          0.634723
age         0.867031
dtype: float64
```

# Exercise 8 – Your own ols() function

## Key Numbers:

- **Custom ols() output:** Perfectly matches the sm.OLS summary.
- **Beta coefficients, Standard Errors, t-statistics, and P-values:** Identical down to the 4th decimal place.
- **Goodness of fit:** $R^2$ and F-statistic computed manually equal the statsmodels results.

## Interpretation:

- By manually implementing the normal equations ($\hat{\beta} = (X^\top X)^{-1} X^\top y$) and calculating the variance of the residuals using pure NumPy, the custom function successfully reproduced the professional statsmodels output. This numerically validates the underlying mathematical theory of least squares estimation, standard errors, and hypothesis testing in multiple linear regression.

```
--- Custom ols() output ---
R^2: 0.7343, F-statistic: 113.5438
       coef  std err        t   P>|t|
0   41.6173   4.9360   8.4313  0.0000
1   -0.1214   0.0330  -3.6784  0.0003
2    0.0470   0.0139   3.3838  0.0008
3    0.0135   0.0621   0.2167  0.8285
4    2.8400   0.8700   3.2643  0.0012
5  -18.7580   3.8514  -4.8705  0.0000
6    3.6581   0.4202   8.7047  0.0000
7    0.0036   0.0133   0.2709  0.7866
8   -1.4908   0.2016  -7.3938  0.0000
9    0.2894   0.0669   4.3254  0.0000
10  -0.0127   0.0038  -3.3365  0.0009
11  -0.9375   0.1322  -7.0914  0.0000
12  -0.5520   0.0507 -10.8968  0.0000

--- statsmodels OLS output ---
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
intercept     41.6173      4.936      8.431      0.000      31.919      51.316
crim          -0.1214      0.033     -3.678      0.000      -0.186      -0.057
zn             0.0470      0.014      3.384      0.001       0.020       0.074
indus          0.0135      0.062      0.217      0.829      -0.109       0.136
...
tax           -0.0127      0.004     -3.337      0.001      -0.020      -0.005
ptratio       -0.9375      0.132     -7.091      0.000      -1.197      -0.678
lstat         -0.5520      0.051    -10.897      0.000      -0.652      -0.452
==============================================================================
```

# Exercise 9 – Regression through the origin

**Key Numbers:**

- **t-statistic ($y$ on $x$ without intercept):** 16.898417
- **t-statistic ($x$ on $y$ without intercept):** 16.898417

**Interpretation:**

- **Symmetry of the t-statistic:** Both models yield the exact same t-statistic. In a regression through the origin (where the intercept is forced to be 0), the t-statistic for testing $H_0$: $\beta = 0$ simplifies algebraically to a symmetric expression:

$$t = \frac{\Sigma x_i y_i}{\sqrt{\Sigma x_i^2 \Sigma y_i^2 - (\Sigma x_i y_i)^2}} \times \sqrt{n - 1}$$

- Because $x$ and $y$ appear symmetrically in this formula, the statistical significance of the linear association between them remains perfectly identical regardless of which variable is treated as the predictor and which as the response.